

III. CGMPs AND THE CONCEPTS OF MODERN QUALITY SYSTEMS

Several key concepts are critical for any discussion of modern quality systems. The following concepts are used throughout this guidance as they relate to the manufacture of pharmaceutical products.

A. Quality Every pharmaceutical product has established identity, strength, purity, and other quality characteristics designed to ensure the required levels of safety and effectiveness. For the purposes of this guidance document, the phrase achieving quality means achieving these characteristics for a product.

B. Quality by Design and Product Development Quality by design means designing and developing a product and associated manufacturing processes that will be used during product development to ensure that the product consistently attains a predefined quality at the end of the manufacturing process.⁵ Quality by design, in conjunction with a quality system, provides a sound framework for the transfer of product knowledge and process understanding from drug development to the commercial manufacturing processes and for post-development changes and optimization. The CGMP regulations, when viewed in their entirety, incorporate the concept of quality by design. This guidance describes how these elements fit together.

C. Quality Risk Management Quality risk management is a valuable component of an effective quality systems framework.⁶ Quality risk management can, for example, help guide the setting of specifications and process parameters for drug manufacturing, assess and mitigate the risk of changing a process or specification, and determine the extent of discrepancy investigations and corrective actions.

D. CAPA (Corrective and Preventive Action) CAPA is a well-known CGMP regulatory concept that focuses on investigating, understanding, and correcting discrepancies while attempting to prevent their recurrence. Quality system models discuss CAPA as three separate concepts, all of which are used in this guidance.

- Remedial corrections of an identified problem
- Root cause analysis with corrective action to help understand the cause of the deviation and potentially prevent recurrence of a similar problem
- Preventive action to avert recurrence of a similar potential problem

E. Change Control Change control is another well-known CGMP concept that focuses on managing change to prevent unintended consequences. The CGMP regulations provide for change control primarily through the assigned responsibilities of the quality control unit. Certain major manufacturing changes (e.g., changes that alter specifications, a critical product attribute or bioavailability) require regulatory filings and prior regulatory approval (21 CFR 314.70, 514.8, and 601.12). Effective change control activities (e.g., quality planning and control of revisions to specifications, process parameters, procedures) are key components of any quality system. In this guidance, change is discussed in terms of creating a regulatory environment that encourages change towards continual improvement. This means a manufacturer is empowered to make changes subject to the regulations based on the

variability of materials used in manufacturing and process improvements resulting from knowledge gained during a product's lifecycle.

F. The Quality Unit Many of the modern quality system concepts described here correlate very closely with the CGMP regulations (refer to the charts later in the document). Current industry practice generally divides the responsibilities of the quality control unit (QCU), as defined in the CGMP regulations, between quality control (QC) and quality assurance (QA) functions.

- QC usually involves (1) assessing the suitability of incoming components, containers, closures, labeling, in-process materials, and the finished products; (2) evaluating the performance of the manufacturing process to ensure adherence to proper specifications and limits; and (3) determining the acceptability of each batch for release.
- QA primarily involves (1) review and approval of all procedures related to production and maintenance, (2) review of associated records, and (3) auditing and performing/evaluating trend analyses.

This guidance uses the term quality unit⁷ (QU) to reflect modern practice while remaining consistent with the CGMP definition in § 210.3(b)(15). The concept of a quality unit is also consistent with modern quality systems in ensuring that the various operations associated with all systems are appropriately planned, approved, conducted, and monitored. The CGMP regulations specifically assign the QU the authority to create, monitor, and implement a quality system. Such activities do not substitute for, or preclude, the daily responsibility of manufacturing personnel to build quality into the product. The QU should not take on the responsibilities of other units of a manufacturer's organization, such as the responsibilities handled by manufacturing personnel, engineers, and development scientists.⁸

Manufacturing personnel and the QU are both critical in fulfilling the manufacturer's responsibility to produce quality products. Other CGMP assigned responsibilities of the QU are consistent with modern quality system approaches (§ 211.22):

- Ensuring that controls are implemented and completed satisfactorily during manufacturing operations
- Ensuring that developed procedures and specifications are appropriate and followed, including those used by a firm under contract to the manufacturer
- Approving or rejecting incoming materials, in-process materials, and drug products
- Reviewing production records and investigating any unexplained discrepancies

Under a quality system, it is normally expected that the product and process development units, the manufacturing units, and the QU will remain independent. In very limited circumstances, a single individual can perform both production and quality functions. That person is still accountable for implementing all the controls and reviewing the results of manufacture to ensure that product quality standards have been met. Under such circumstances, it is recommended that another qualified individual, not involved in the production operation, conduct an additional, periodic review of QU activities.

G. Six-system Inspection Model The FDA's Drug Manufacturing Inspection Compliance Program, which contains instructions to FDA personnel for conducting inspections, is a systems-

based approach to inspection and is very consistent with the robust quality system model presented in this guidance.⁹ The diagram below shows the relationship among the six systems: the quality system and the five manufacturing systems. The quality system provides the foundation for the manufacturing systems that are linked and function within it. The quality system model described in this guidance does not consider the five manufacturing systems as discrete entities, but instead integrates them into appropriate sections of the model. Those familiar with the six-system inspection approach will see organizational differences in this guidance; however, the inter-relationship should be readily apparent. One of the important themes of the systems based inspection compliance program is that you have the ability to assess whether each of the systems is in a state of control. The quality system model presented in this guidance will also serve to help firms achieve this state of control.

IV THE QUALITY SYSTEMS MODEL

C. Manufacturing

Significant overlap exists between the elements of a quality system and the CGMP regulation requirements for manufacturing operations. It is important to emphasize again that FDA's enforcement programs and inspectional coverage remain based on the CGMP regulations. When quality system elements in this section do not correlate to the CGMP regulations, the guidance makes recommendations to help facilitate compliance with the CGMP regulations. The language in this section has been tailored to the pharmaceutical manufacturing environment.

1. Design, Develop, and Document Product and Processes

In a modern quality systems manufacturing environment, the significant characteristics of the product being manufactured should be defined from design to delivery, and control should be exercised over all changes. In addition, quality and manufacturing processes and procedures — and changes to them — must be defined, approved, and controlled (§ 211.100). It is important to establish responsibility for designing or changing products. Documenting processes, associated controls, and changes to these processes will help ensure that sources of variability are identified. Documentation includes:

- Resources and facilities used
- Procedures to carry out the process
- Identification of the process owner who will maintain and update the process as needed
- Identification and control of important variables
- Quality control measures, necessary data collection, monitoring, and appropriate controls for the product and process
- Any validation activities, including operating ranges and acceptance criteria
- Effects on related process, functions, or personnel

As discussed under section IV.A., above, the model calls for managers to ensure that product specifications and process parameters are determined by the

appropriate technical experts (e.g., engineers, development scientists). In the pharmaceutical environment, experts would have an understanding of pharmaceutical science, equipment, facilities, and process types and of how variations in materials and processes can ultimately affect the finished product. Packaging and labeling controls, critical stages in the pharmaceutical manufacturing process, are not specifically addressed in quality systems models. However, the Agency recommends that manufacturers always refer to the packaging and labeling control regulations at § 211 Subpart G. In addition — and this is consistent with modern quality systems — FDA recommends that, as part of the design process, before commercial production, the controls for all processes within the packaging and labeling system be planned and documented in written procedures. The procedures should outline quality control activities and the responsible positions. Specifications and controls for the packaging and labeling materials should also be determined before commercial production. Distinct labels with discriminating features for different products, such as a product marketed with different strengths, should be included to prevent mislabeling and resulting recalls.

2. Examine Inputs In a modern quality systems model, the term input includes any material that goes into a final product, no matter whether the material is purchased by the manufacturer or produced by the manufacturer for the purpose of processing. Materials can include items such as components (e.g., ingredients, process water, and gas), containers, and closures. A robust quality system will ensure that all inputs to the manufacturing process are reliable because quality controls will have been established for the receipt, production, storage, and use of all inputs. The CGMP regulations require either testing or use of a certificate of analysis (COA) plus an identity analysis (§ 211.84) for the release of materials for manufacturing. In the preamble to the CGMP regulations, these requirements were explicitly interpreted.¹² The preamble states that reliability can be validated by conducting tests or examinations and comparing the results to the supplier's COA. Sufficient initial tests should be done to establish reliability and to determine a schedule for periodic reassessment. As an essential element of purchasing controls, it is recommended that data trends for acceptance and rejection of materials be analyzed for information on supplier performance.¹³ The quality systems approach also calls for periodic auditing of suppliers based on risk assessment. During the audit, a manufacturer can observe the testing or examinations conducted by the supplier to help determine the reliability of the supplier's COA. An audit should also include a systematic examination of the supplier's quality system to ensure that reliability is maintained. It is recommended that a combination approach be used (i.e., verify the suppliers' COA through analysis and audits of the supplier). Under a quality systems approach, if full analytical testing is not done, the audit should cover the supplier's analysis (i.e., a specific identity test is still

required under § 211.84(d)(2)). Under a quality systems approach, procedures should be established to verify that materials are from qualified sources (for application and licensed products, certain sources are specified in the submissions). Procedures should also be established to encompass the acceptance, use, or the rejection and disposition of materials produced by the facility (e.g., purified water). Systems that produce these in-house materials should be designed, maintained, qualified, and validated where appropriate to ensure that the materials meet their acceptance criteria. In addition, it is recommended that changes to materials (e.g., specification, supplier, or materials handling) be implemented through a change control system (certain changes require review and approval by the QU (§ 211.100(a))). It is also important to have a system in place to respond to changes in materials from suppliers so that necessary adjustments to the process can be made and unintended consequences avoided.

3. Perform and Monitor Operations An important purpose of implementing a quality systems approach is to enable a manufacturer to more efficiently and effectively validate, perform, and monitor operations (§ 211.100(a)) and ensure that the controls are scientifically sound and appropriate. The goal of establishing, adhering to, measuring, and documenting specifications and process parameters is to objectively assess whether an operation is meeting its design and product performance objectives. In a robust quality system, production and process controls should be designed to ensure that the finished products have the identity, strength, quality, and purity they purport or are represented to possess (see, e.g., § 211.100(a)). In a modern quality system, a design concept established during product development typically matures into a commercial design after process experimentation and progressive modification. Risk management can help identify areas of process weakness or higher risk and factors that can influence critical quality attributes that should receive increased scrutiny. The FDA recommends that scale-up studies be used to help demonstrate that a fundamentally sound design has been fully realized. A sufficiently robust manufacturing process should be in place prior to commercial production. With proper design (see IV.C.1.) and reliable mechanisms to transfer process knowledge from development to commercial production, a manufacturer should be able to validate the manufacturing process.¹⁴ Conformance batches provide initial proof that the design of the process produces the intended product quality. Sufficient testing data will provide essential information on performance of the new process, as well as a mechanism for continual improvement. Modern equipment with the potential for continual monitoring and control can further enhance this knowledge base. Although initial commercial batches can provide evidence to support the validity and consistency of the process,¹⁵ the entire product life cycle should be addressed by the establishment of continual improvement mechanisms in the quality system.¹⁶ Thus, in accordance with the

quality systems approach, process validation is not a one-time event, but an activity that continues throughout a product's life. As experience is gained in commercial production, opportunities for process improvements may become evident. (CGMP regulations § 211.180 require the review and evaluation of records to determine the need for any change. These records contain data and information from production that provide insight into the product's state of control. Change control systems should provide a In addition, it is recommended that changes to materials (e.g., specification, supplier, or materials handling) be implemented through a change control system (certain changes require review and approval by the QU (§ 211.100(a)). It is also important to have a system in place to respond to changes in materials from suppliers so that necessary adjustments to the process can be made and unintended consequences avoided.

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opportunities for process improvements may become evident. (CGMP regulations § 211.180 require the review and evaluation of records to determine the need for any change. These records contain data and information from production that provide insight into the product's state of control. Change control systems should provide a dependable mechanism for prompt implementation of technically sound manufacturing improvements.) Under a quality system, written procedures are followed and deviations from them are justified and documented (CGMP requires this; see § 211.100(b)) to ensure that the manufacturer can trace the history of the product, as appropriate, concerning personnel, materials, equipment, and chronology and that processes for product release are complete and recorded. Both the CGMP regulations (§ 211.110) and quality systems models call for the monitoring of critical processes that may be responsible for causing variability during production. For example:

- Process steps must be verified by a second person (§ 211.188).
- Process steps can also be performed using a validated computer system.

Batch production records must be prepared contemporaneously with each phase of production (§ 211.100(b)). Although time limits for production can be established when they are important to the quality of the finished product (CGMP addresses this; see § 211.111), the manufacturer should have the ability to establish production controls using in-process parameters that are based on desired process endpoints measured using real time testing or monitoring apparatus (e.g., blend until mixed vs. blend for 10 minutes). Procedures must be in place to prevent objectionable microorganisms in finished products not required to be sterile and to prevent microbial contamination of finished products purported to be sterile. Sterilization processes must be validated for sterile drugs (§ 211.113(b)).¹⁷ Manufacturing processes must consistently meet their parameters, and in-process materials must meet acceptance criteria or limits (§ 211.110(b) and (c)) so that, ultimately, finished pharmaceutical products will meet their acceptance criteria. Under a quality system, selected data are used to evaluate the quality of a process or product. In addition, data collection can provide a means to encourage and analyze potential suggestions for improvement. A quality systems approach calls for the manufacturer to develop procedures that monitor, measure, and analyze the operations (including analytical methods and/or statistical techniques). Monitoring of the process is important due to the limitations of testing. Knowledge continues to accumulate from development through the entire commercial life of a product. Significant unanticipated variables should be detected by a well-managed quality system and adjustments implemented. Procedures should be revisited as needed to refine operational design based on new knowledge. Process understanding increases with experience and helps identify when change will lead to continual improvement. When implementing data collection procedures, consider the following:

- Are data collection methods documented? When in the product life cycle will the data be collected? How and to whom will measurement and monitoring activities

be assigned? When should analysis and evaluation (e.g. trending) of laboratory data be performed? (see IV.D.1)

- What records should be collected? A modern quality system approach indicates that change control is warranted when data analysis or other information reveals an area for improvement. Changes to an established process must be controlled and documented to ensure that desired attributes for the finished product will be met (§ 211.100(a)). Change control with regard to pharmaceuticals is addressed in more detail in the CGMP regulations. When developing a process change, it is important to keep the process design and scientific knowledge of the product in mind. If major design issues are encountered through process experience, a firm may want to revisit the adequacy of the design of the manufacturing facility (§ 211.42), the design of the manufacturing equipment (§ 211.63), the design of the production and control procedures (§ 211.100), or the design of laboratory controls (§ 211.160). When implementing a change, its effect should be determined by monitoring and evaluating those specific elements that may be affected based on an understanding of the process. This approach allows the steps taken to implement a change and the effects of the change on the process to be considered systematically. Application of risk analysis may facilitate evaluating the potential effect of the change. Evaluating the effects of a change can entail additional tests or examinations of subsequent batches (e.g., additional in-process testing or additional stability studies). The quality system elements identified in this guidance, if implemented and maintained, will help a manufacturer manage change and implement continual improvement in manufacturing. Under a quality systems approach, procedures should be in place to ensure the accuracy of test results. Test results that are out of specification may be due to testing problems or manufacturing problems and should be investigated. Any invalidation of a test result should be scientifically sound and justified. To maintain quality, the Agency recommends that prior to completion of manufacturing, the manufacturer should consider storage and shipment requirements to meet special handling needs (in the case of pharmaceuticals, one example might be refrigeration). Under a quality system, trends should be continually identified and evaluated. One way of accomplishing this is the use of statistical process control. The information from trend analyses can be used to continually monitor quality, identify potential variances before they become problems, bolster data already collected for the annual review, and facilitate improvement throughout the product life cycle. Process capability assessment can serve as a basis for determining the need for changes that can result in process improvements and efficiency (see IV.D.1.).

D. Evaluation Activities

As in the previous section, the elements of a quality system correlate closely with the requirements in the CGMP regulations. See the table at the end of the section for the specifics.

1. **Analyze Data for Trends** Quality systems call for continually monitoring trends and improving systems. This can be achieved by monitoring data and information, identifying and resolving problems, and anticipating and preventing problems. Quality systems procedures involve collecting data from monitoring, measurement, complaint handling, or other activities, and tracking this data over time, as appropriate. Analysis of data can provide indications that controls are losing effectiveness. The information generated will be essential to achieving problem resolution or problem prevention (see IV.D.3.). Although the CGMP regulations (§ 211.180(e)) require product review on at least an annual basis, a quality systems approach calls for trending on a more frequent basis as determined by risk. Trending enables the detection of potential problems as early as possible to plan corrective and preventive actions. Another important concept of modern quality systems is the use of trending to examine processes as a whole; this is consistent with the annual review approach. Trending analyses can help focus internal audits (see IV.D.2.).

2. **Conduct Internal Audits** A quality systems approach calls for audits to be conducted at planned intervals to evaluate effective implementation and maintenance of the quality system and to determine if processes and products meet established parameters and specifications. As with other procedures, audit procedures should be developed and documented to ensure that the planned audit schedule takes into account the relative risks of the various quality system activities, the results of previous audits and corrective actions, and the need to audit the complete system. Procedures should describe how auditors are trained in objective evidence gathering, their responsibilities, and auditing procedures. Procedures should also define auditing activities such as the scope and methodology of the audit, selection of auditors, and audit conduct (audit plans, opening meetings, interviews, closing meeting and reports). It is critical to maintain records of audit findings and assign responsibility for follow-up to prevent problems from recurring (see IV.D.3.). The quality systems model calls for managers who are responsible for the areas audited to take timely action to resolve audit findings and ensure that follow-up actions are completed, verified, and recorded. (FDA's policy is to refrain from both reviewing and copying reports or records that result from internal audits per Compliance Policy Guide 130.300.19)

3. **Quality Risk Management** Effective decision-making in a quality systems environment is based on an informed understanding of quality issues. Elements of risk should be considered relative to intended use of a product, and in the case of pharmaceuticals,

patient safety and ensuring availability of medically necessary drug products. Management should assign priorities to activities or actions based on an assessment of the risk including both the probability of occurrence of harm and of the severity of that harm. It is important to engage appropriate parties in assessing the risk. Such parties include customers, appropriate manufacturing personnel, and other stakeholders.

Implementation of quality risk management includes assessing the risks, selecting and implementing risk management controls commensurate with the level of risk, and evaluating the results of the risk management efforts. Since risk management is an iterative process, it should be repeated if new information is developed that changes the need for, or nature of, risk management. In a manufacturing quality systems environment, risk management is used as a tool in the development of product specifications and critical process parameters. Used in conjunction with process understanding, quality risk management helps manage and control change.

4. Corrective Action Corrective action is a reactive tool for system improvement to ensure that significant problems do not recur. Both quality systems and the CGMP regulations emphasize corrective actions. Quality systems approaches call for procedures to be developed and documented to ensure that the need for action is evaluated relevant to the possible consequences, the root cause of the problem is investigated, possible actions are determined, a selected action is taken within a defined timeframe, and the effectiveness of the action taken is evaluated. It is essential to document corrective actions taken (CGMP also requires this; see § 211.192). It is essential to determine what actions will reduce the likelihood of a problem recurring. Examples of sources that can be used to gather such information include the following:

- Contains Nonbinding Recommendations • • • • •
- Nonconformance reports and rejections
- Returns
- Complaints
- Internal and external audits
- Data and risk assessment related to operations and quality system processes
- Management review decisions

5. Preventive Actions Being proactive is an essential tool in quality systems management. Succession planning, training, capturing institutional knowledge, and planning for personnel, policy, and process changes are preventive actions that will help ensure that potential problems and root causes are identified, possible consequences assessed, and appropriate actions considered. The selected preventive action should be evaluated and recorded, and the system should be monitored for the effectiveness of the action. Problems can be anticipated and their occurrence prevented by reviewing data and analyzing risks associated with operational and quality system processes, and by keeping abreast of changes in scientific developments and regulatory requirements.

6. Promote Improvement The effectiveness and efficiency of a quality system can be improved through the quality activities described in this guidance. Management may choose to use other improvement activities as appropriate. It is critical that senior management be involved in the evaluation of this improvement process (see IV.D.3.). The following table shows how the CGMP regulations correlate to specific elements in the quality systems model for this section. Manufacturers should always refer to the specific regulations to make sure they are complying with all regulations.